Epidemiological Patterns of Initial and Subsequent Injuries in Collegiate Football Athletes

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Background: A body of epidemiological studies has examined football injuries and associated risk factors among collegiate athletes. However, few existing studies specifically analyzed injury risk in terms of initial or subsequent injuries.

Purpose: To determine athlete-exposures (AEs) and rates of initial and subsequent injury among collegiate football athletes.

Study Design: Descriptive epidemiological study.

Methods: Injury and exposure data collected from collegiate football players from two Division I universities (2007-2011) were analyzed. Rate of initial injury was calculated as the number of initial injuries divided by the total number of AEs for initial injuries, while the rate for subsequent injury was calculated as the number of subsequent injuries divided by the total number of AEs for subsequent injury. Poisson regression was used to determine injury rate ratio (subsequent vs initial injury), with adjustment for other covariates.

Results: The total AEs during the study period were 67,564, resulting in an overall injury rate of 35.2 per 10,000 AEs. Rates for initial and subsequent injuries were 31.7 and 45.3 per 10,000 AEs, respectively, with a rate ratio (RR) of 1.4 for rate of subsequent injury vs rate of initial injury (95% CI, 1.1-1.9). Rate of injury appeared to increase with each successive injury. RR during games was 1.8 (95% CI, 1.1-3.0). The rate of subsequent injuries to the head, neck, and face was 10.9 per 10,000 AEs, nearly double the rate of initial injuries to the same sites (RR = 2.0; 95% CI, 1.1-3.5). For wide receivers, the rate of subsequent injuries was 2.2 times the rate of initial injuries (95% CI, 1.3-3.8), and for defensive linemen, the rate of subsequent injuries was 2.1 times the rate of initial injuries (95% CI, 1.1-3.9).

Conclusion: The method used in this study allows for a more accurate determination of injury risk among football players who have already been injured at least once. Further research is warranted to better identify which specific factors contribute to this increased risk for subsequent injury.

Keywords: football (American); epidemiology; collegiate athletics; subsequent injury; recurrent injury

Collegiate athletics continue to increase in popularity, and rates of participation are increasing. However, despite improvements in medical attention and changes in rules, collegiate athletes maintain exceptionally high rates of injuries.[2,10,11] Football, in particular, has one of the highest reported injury rates of any sport, with a reported injury rate of 9.6 injuries per 1000 athlete-exposures (AEs) for practices and a rate of 35.9 injuries per 1000 AEs for games.[3,10] While several epidemiological studies have examined football injuries and associated risk factors among collegiate athletes,[2,3,10,21] none of these specifically analyzed these injuries as initial or subsequent injuries.[14] The few previous studies examining initial or subsequent injuries focused primarily on high school populations[1,18,23] or were limited to specific types of injuries.[5,6]

Existing research shows that athletes with a history of injury are more likely to experience another injury,[5,6,8,13,23] and each successive injury can further increase the risk of another subsequent injury.[14] Prior studies have also found that subsequent injuries are often associated with more severe symptoms and/or consequences than initial injuries.[14,18,23] For some types of subsequent injuries, athletes are more likely to require surgery before returning to play or to stop athletic participation altogether.[19,23] However, while subsequent injury is an important issue that requires additional research, estimating the risk of subsequent injuries has been challenging, mainly due to difficulties in documenting and tracking AEs for subsequent injuries. Most existing studies do not distinguish exposures for initial injuries from those of subsequent injuries. The total number of AEs is often used as a denominator when calculating the rate for both initial injuries and subsequent injuries.[1,23] Such an approach is likely to result in an underestimation of the risk of subsequent injuries because athletes are not at risk of a subsequent injury until after they have sustained

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an initial injury.[14] Other existing studies used the proportion of athletes injured or simply used history of injury as a proxy when estimating the risk of subsequent injuries, which could also bias the results because all athletes do not have the same AEs.[14,18,20,23] To accurately determine the risk of subsequent injuries, it is important to better quantify AEs among the individuals who are at risk for a subsequent injury. The purpose of this study was to determine rates of initial and subsequent injury using the corresponding exposures and to compare rates of initial and subsequent football injuries among Division I collegiate athletes. Understanding the risk of subsequent injuries is the first step to preventing these injuries.

METHODS

Study Population and Data Collection

Data for this study came from a prospective cohort study conducted among athletes playing 9 team sports at three Division I universities from 2007 to 2012. Details about the study have been described elsewhere.[24-26] For purposes of this study, data collected from collegiate football players on the rosters of two Division I universities from 2007 to 2011 were used. Players who were at least 18 years of age, participated in football during the study period, and signed informed consent were included. During the first year of the study, athletes from all grades were enrolled. In subsequent years, only new freshmen and transfers joining the team were eligible for enrollment. The participation rate was 87% at the athlete level.

Participating football players completed baseline surveys upon enrollment and were followed prospectively for injury during the study. Team athletic trainers reported injuries through the Sports Injury Monitoring System (SIMS), an existing injury surveillance system that collects injury and exposure information. If an injury met the study inclusion criteria, the injured player then completed follow-up surveys at multiple time intervals (eg, 1 week, 1 month, 3 months, 6 months, and 12 months) until return to play. The final return-to-play survey was completed within 1 week after the injured player returned to play, regardless of the time missed due to the injury. For this study, data from the baseline surveys, the existing injury database, 1-week follow-up surveys, and return-to-play survey were merged for analysis. The institutional review board (IRB) of the authors' institutions approved this study.

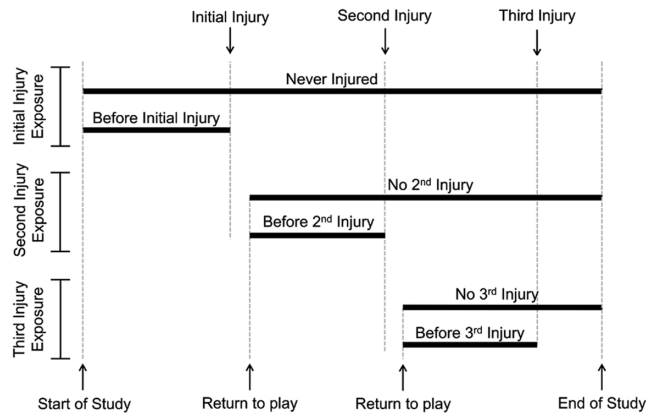

Figure 1. Breakdown of athlete-exposure by injury number. The thicker lines indicate the entire time a player is at risk of the next injury.

Initial and Subsequent Injury

Injury was defined as an event that occurred during football games or practices, required medical attention, and resulted in loss of play for at least 1 day. Each injury was recorded either as an "initial injury" if it was the athlete's first reportable injury during the study period or as a "subsequent injury" if the athlete sustained at least 1 prior injury during the study period, regardless of injury type or injury site.

Athlete-Exposure

One AE was defined as attending 1 coach-directed session (either game or practice). To document the risk for each injury, AEs were calculated for each injury. Specifically, the AEs for an initial injury were counted from when the athlete was enrolled in the study and then continued until the athlete's first injury or until the end of the study if the athlete was never injured (Figure 1). Total AEs for each initial injury were calculated as the sum of coach-directed sessions for each athlete before the initial injury, as total game or practice exposures, respectively. The AEs for subsequent injury were calculated only among athletes who had an initial injury during the study period. Since there were 0 to 6 injuries per athlete, subsequent injuries for 1 player can be up to 5 in this study. Thus, the total AEs for subsequent injuries were calculated by adding the AEs for each subsequent injury. Specifically, the total AEs for each second injury were calculated as the sum of AEs occurring between the

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time when the injured athlete returned to play after initial injury and when the second injury occurred, or until the end of the study if the player was not injured again. The AEs for all subsequent injuries (third, fourth, etc) followed the same method of calculation. In addition, some athletes in their senior year only practiced until the last game, so their exposures were only counted until the date of last game.

Rate of Injury

Risk of initial or subsequent injury was measured as respective injury rates. Rate of initial injury was calculated as the number of initial injuries sustained during the study period, divided by the total number of AEs for initial injuries in the same period, multiplied by 10,000. Rate of subsequent injury was calculated as the number of subsequent injuries sustained during the study period divided by the total number of AEs for subsequent injury (AEs for the second through sixth injuries combined), multiplied by 10,000. The initial game injury rate was calculated as the number of initial injuries sustained in games during the study period, divided by the total number of AEs for initial injuries in the same period, multiplied by 10,000. This method was also used to calculate rates for initial practice, subsequent game, and subsequent practice injuries.

Other Covariates

Other covariates included demographic information (ie, age, grade, university, body mass index), football position, years of experience (defined as number of years played before enrollment in study), and history of injury (defined as an injury sustained within 12 months before the injury). In addition, detailed information about each reportable injury (ie, type, location, mechanism, and severity of injury) and each AE (ie, date of each practice and game activity) was included. Given a lack of standard terminology and the small number of recurrent injuries (injury of the same type to the same body part) (n = 6), we included both recurrent injuries and other forms of subsequent injury (ie, different injury types and/or body locations) in our definition of subsequent injury.

Statistical Analysis

Participant demographic and injury characteristics were described. Differences in proportions between initial and subsequent injuries across participant demographic characteristics were compared by means of Pearson chi-square tests. Poisson regression models with log-link function were used to calculate the rates of initial and subsequent injuries, along with 95% confidence intervals, during games, practices, and both. Finally, adjusted Poisson regression models were used to examine the factors associated with increased rate of subsequent injuries versus initial injuries. A P value of $<.05$ was considered statistically significant, and all analyses were conducted by use of SAS 9.3 (SAS Institute).

RESULTS

Demographics

During the 4-year study, a total of 365 football players from 2 universities were enrolled, with 173 from university A and 192 from university B. A total of 238 reportable injuries were sustained by 159 players during the study period. Of the 238 injuries, 67% were initial injuries (n = 159) and 33% were subsequent injuries (n = 79) (Table 1). A total of 20 concussions were reported in this study, including 2 recurrent concussions. No significant differences were found in the proportion of initial and subsequent injuries between the 2 universities. However, being a college senior or having older age at enrollment was associated with high proportions of reported subsequent injury. Having a history of injury in the 12 months before injury was also significantly associated with subsequent injury.

Injury Rate

The total AEs during the study period were 67,564, resulting in an overall injury rate of 35.2 per 10,000 AEs (Table 2). Although the number of initial injuries (n = 159) was greater than the number of subsequent injuries (n = 79), the rate of subsequent injuries was 1.4 (rate ratio [RR] = 1.4; 95% CI, 1.1-1.9) times higher than that of initial injuries, with a rate of 45.3 per 10,000 AEs for subsequent injuries and 31.7 per 10,000 AEs for initial injuries. Although fewer injuries occurred during games (n = 64) than practices (n = 174), the injury rate during games was 97.7 per 10,000 AEs (95% CI, 76.5-124.8), which was more than 3 times higher than the injury rate during practices (28.5 per 10,000 AEs; 95% CI, 24.6-33.1). In both game and practice settings, the subsequent injury rate was higher than the initial injury rate. However, the difference was significant only in games, where the rate of subsequent injury was 1.8 times the rate of initial injuries (RR = 1.8; 95% CI, 1.1-3.0). Overall, the injury rate increased with each subsequent injury until the fourth injury, after which the rate began to decrease but remained higher than the rate of initial injury. Specifically, the rates for second, third, fourth, fifth, and sixth injury were 41.1, 77.5, 151.5, 116.7, and 64.9 per 10,000 AEs, respectively.

The rate of subsequent injuries to the head, neck, and face was 10.9 per 10,000 AEs, which was double the rate of initial injuries to the same sites (RR = 2.0; 95% CI, 1.1-3.5) (Table 3). The majority of the subsequent injuries to the head, neck, and face were concussions (63%; RR = 1.7; 95% CI, 0.8-3.5). The most common types of initial injuries were sprains, with a rate of 14.6 per 10,000 AEs, followed by strains (5.4 per 10,000 AEs) and concussions (4.0 per 10,000 AEs).

Wide receivers and defensive linemen also had significantly higher rates of subsequent injuries compared with initial injuries. For wide receivers, the rate of subsequent injuries was 2.2 times the rate of initial injuries (95% CI, 1.3-3.8), and for defensive linemen, the rate of subsequent injuries was 2.1 times the rate of initial injuries (95% CI, 1.1-3.9). Many other players also showed higher rates for subsequent injuries, although the differences were not statistically significant, due, in part, to small sample size (Table 3).

TABLE 1
Characteristics of Injured Participants at Enrollment in the Study

Characteristic	Injuries, n (%)			
	All (n = 238)	Initial (n = 159)	Subsequent (n = 79)	*P* Value
Age, y				
18	76 (32.2)	66 (41.8)	10 (12.8)	<.0001[a]
19-20	101 (42.8)	67 (42.4)	34 (43.6)	
21-22	51 (21.6)	23 (14.6)	28 (35.9)	
23-24	8 (3.4)	2 (1.3)	6 (7.7)	
University				
A	130 (54.6)	88 (55.4)	42 (53.2)	.7503
B	108 (45.4)	71 (44.7)	37 (46.8)	
Grade				
Freshman[b]	114 (47.9)	103 (64.8)	11 (13.9)	<.0001[a]
Sophomore	50 (21.0)	29 (18.2)	21 (26.6)	
Junior	37 (15.5)	16 (10.1)	21 (26.6)	
Senior	37 (15.5)	11 (6.9)	26 (32.9)	
Years of experience				
<5 y	53 (29.3)	39 (31.5)	14 (24.6)	.0935
5-6 y	97 (53.6)	60 (48.4)	37 (64.9)	
>6 y	31 (17.1)	25 (20.2)	6 (10.5)	
12-month history of injury[c]				
Yes	92 (38.7)	33 (20.8)	59 (74.7)	<.0001[a]
No	146 (61.3)	126 (79.3)	20 (25.3)	

[a]Indicates significance with $P < .05$.
[b]"Freshman" includes true freshmen and red shirt freshmen (indicates academic second year using first year of eligibility for athletic participation).
[c]"History of injury" reflects status at time of injury. All other variables reflect status at time of enrollment.

TABLE 2
Rates of Initial and Subsequent Injury per 10,000 Athlete-Exposures (n = 238 injuries)[a]

Classification	All Injuries				Initial Injury				Subsequent Injury					
	n[b]	AEs	Rate	95% CI	n[b]	AEs	Rate	95% CI	n[b]	AEs	Rate	95% CI	RR[c]	95% CI
Overall	238	67,564	35.2	31.0-40.0	159	50,118	31.7	27.2-37.1	79	17,446	45.3	40.2-62.5	1.4	1.1-1.9
Game	64	6551	97.7	76.5-124.8	38	4747	80.1	58.3-110.0	26	1804	144.1	98.1-211.7	1.8	1.1-3.0
Practice	174	61,013	28.5	24.6-33.1	121	45,371	26.7	22.3-31.9	53	15,642	33.9	30.3-38.0	1.3	0.9-1.8

[a]AE, athlete-exposure; CI, confidence interval; RR, injury rate ratio.
[b]Number of injuries.
[c]Initial injury as reference group.

Risk Factors for Second Injuries

The highest proportions of second injuries were among athletes who initially had dislocations (50.0%), fractures (41.7%), and concussions (40.0%) (Figure 2). The most common body site for initial injuries was lower extremities (52.2%), and athletes with initial lower extremity injuries also had the highest percentage sustaining any second injury (37.3%). The majority of athletes with an initial injury had an injury severity ranging from 1 week to 1 month loss of play. However, the proportion of athletes who sustained a second injury was highest among those who lost 3 months of play after their first injury (50.0%). Athlete position also affected the injury rate, with the highest rate among running backs (45.9 per 10,000 AEs), followed by defensive backs (35.0 per 10,000 AEs) and offensive linemen (34.0 per 10,000 AEs) (Table 3). However, the positions with the greatest proportion of initial injuries continuing to second injuries were defensive linemen (52%), offensive linemen (37.9%), and receivers (31.8%) (Figure 2).

DISCUSSION

This study described the rates of initial and subsequent injuries among collegiate football players and examined factors associated with increased risk of subsequent injuries compared with initial injuries. The main findings showed that overall, the rate of subsequent injuries was nearly 1.5 times the rate of initial injuries. Rate of injury also appeared to

TABLE 3
Initial and Subsequent Injury Rates Per 10,000 Athlete-Exposures During Games and Practices[a]

	Game						Practice						Total					
	Initial		Subsequent				Initial		Subsequent				Initial		Subsequent			
Characteristic	n[b]	Rate	n[b]	Rate	RR[c]	95% CI	n[b]	Rate	n[b]	Rate	RR[c]	95% CI	n[b]	Rate	n[b]	Rate	RR[c]	95% CI
All injuries	38	80.1	26	144.1	1.8	1.1-3.0[d]	121	26.7	53	33.9	1.3	0.9-1.8	159	31.7	79	45.3	1.4	1.1-1.9[d]
Body region																		
Head, neck, face	7	14.8	7	38.8	2.6	0.9-7.5	21	4.6	12	7.7	1.7	0.8-3.4	28	5.6	19	10.9	2.0	1.1-3.5[d]
Upper extremity	10	21.1	5	27.7	1.3	0.5-3.9	20	4.4	9	5.8	1.3	0.6-2.9	30	6.0	14	8.0	1.3	0.7-2.5
Trunk	2	4.2	1	5.5	1.3	0.2-14.5	14	3.1	3	1.9	0.6	0.2-2.2	16	3.2	4	2.3	0.7	0.2-2.2
Lower extremity	19	40.0	13	72.1	1.8	0.9-3.7	64	14.1	25	16.0	1.1	0.7-1.8	83	16.6	38	21.8	1.3	0.9-1.9
Other	0	0	0	0	—	—	2	0.4	4	2.6	5.8	1.1-31.7[d]	2	0.4	4	2.3	5.8	1.1-31.4[d]
Type of injury																		
Sprain	24	50.6	10	55.4	1.1	0.5-2.3	49	10.8	19	12.2	1.1	0.7-1.9	73	14.6	29	16.6	1.1	0.7-1.8
Strain	2	4.2	1	5.5	1.3	0.1-14.5	25	5.5	5	3.2	0.9	0.2-1.5	27	5.4	6	3.4	0.6	0.3-1.6
Concussion	5	10.5	5	27.7	2.6	0.8-9.1	15	3.3	7	4.5	1.4	0.6-3.3	20	4.0	12	6.9	1.7	0.8-3.5
Fracture	1	2.1	3	16.6	7.9	0.8-75.9	11	2.4	6	3.8	1.6	0.6-4.3	12	2.4	9	5.2	2.2	0.9-5.1
Contusion-bruise	0	0	1	5.5	—	—	3	0.7	1	0.6	1.0	0.1-9.3	3	0.6	2	1.2	1.9	0.3-11.5
Dislocation	2	4.2	1	5.5	1.3	0.1-14.5	2	0.4	2	1.3	2.9	0.4-20.6	4	0.8	3	1.7	2.2	0.5-9.6
Other	4	8.4	5	27.7	3.3	0.9-12.3	16	3.5	13	8.3	2.4	1.1-4.9[d]	20	4.0	18	10.3	2.6	1.4-4.9[d]
Severity																		
1 wk	14	29.5	9	49.9	1.7	0.7-3.9	56	12.3	23	14.7	1.2	0.7-1.9	69	14.0	32	18.3	1.3	0.9-2.0
1 mo	15	31.6	10	55.4	1.8	0.8-3.9	34	7.5	16	10.2	1.4	0.8-2.5	49	9.8	26	14.9	1.5	0.9-2.5
3 mo	3	6.32	3	16.6	2.6	0.5-13.0	17	3.8	6	3.8	1.0	0.4-2.6	20	4.0	9	5.2	1.3	0.6-2.8
6 mo	4	8.43	3	16.6	2.0	0.4-8.8	11	2.4	5	3.2	1.3	0.5-3.8	15	3.0	8	4.6	1.5	0.7-3.6
>6 mo	2	4.21	1	5.5	1.3	0.1-14.5	3	0.7	3	1.9	2.9	0.6-14.4	5	1.0	4	2.3	2.3	0.6-8.6
Position																		
Offensive lineman	4	49.6	2	59.5	1.2	0.2-6.6	25	32.4	12	49.9	1.5	0.8-3.0	29	34.0	14	51.0	1.5	0.8-2.8
Quarterback	1	49.9	0	0.0	—	—	1	5.2	0	0.0	—	—	2	9.5	0	0.0	—	—
Running back	6	132.3	5	203.1	1.5	0.8-5.0	16	36.9	5	26.3	0.7	0.3-1.9	22	45.9	10	46.5	1.0	0.5-2.1
Wide receiver	8	93.1	7	229.8	2.5	0.9-6.6	22	26.8	12	51.5	1.9	1.0-3.8	30	33.1	19	71.7	2.2	1.3-3.8[d]
Defensive lineman	5	69.2	4	141.9	2.1	0.6-7.6	20	29.0	10	59.5	2.1	1.0-4.1[d]	25	32.8	14	68.9	2.1	1.1-3.9[d]
Linebacker	7	139.7	3	193.9	1.4	0.4-5.4	7	14.6	3	25.0	1.7	0.4-6.6	14	26.5	6	44.3	1.7	0.6-4.4
Defensive back	6	96.4	3	118.5	1.2	0.3-4.9	17	28.6	4	20.5	0.7	0.2-2.1	23	35.0	7	31.8	0.9	0.4-2.1
Other	1	26.7	0	0.0	—	—	5	14.0	0	0.0	—	—	6	15.2	0	0.0	—	—
12-month history of injury																		
Yes	7	47.3	17	214.7	4.5	1.9-11.0	26	18.4	42	68.37	3.7	2.3-6.1	33	21.1	59	85.1	4.0	2.6-6.2
No	31	95.5	9	85.60	0.9	0.4-1.9	95	30.7	11	13.89	0.5	0.2-0.9	126	36.8	20	22.3	0.6	0.4-1.0

[a]— indicates that values could not be calculated given available data. CI, confidence interval; RR, injury rate ratio.
[b]Number of injuries.
[c]Initial injury as reference group.
[d]Indicates significance with $P < .05$.

increase with each successive injury. While the subsequent injury rate was higher than the initial injury rate in both games and practices, the difference was significant only in games, with the rate of subsequent injuries about 1.8 times the rate of initial injuries. While running backs had a higher rate of initial injuries, wide receivers and defensive linemen had significantly higher rates of subsequent injuries compared with initial injuries. Additionally, the rate of subsequent injuries to the head, neck, and face was almost double the rate of initial injuries. More injuries were observed among freshmen because an additional cohort of incoming freshmen were added to the study each year. These results add empirical evidence to existing study findings regarding athletic injuries in college football, whereby once athletes are injured, they are more likely to be injured again. Our results highlight the need to increase efforts to prevent subsequent injuries among collegiate athletes.

This is one of few studies that estimate subsequent injury rates using only the AEs of athletes who already had an initial injury. Most published studies reporting subsequent injury rates do not separate the number of AEs for initial injuries and/or for subsequent injuries[1,23]; however, this distinction is important because only athletes who sustained an initial injury are at risk for a subsequent injury.[14] Using the same number of total exposures for both initial and subsequent injuries could result in an underestimated rate of subsequent injuries due to overestimating the AEs. The method used in this study took into consideration the length of time the athlete was at risk before each injury, allowing for greater accuracy in the capture of the AEs for athletes who are at risk for a subsequent injury. This method also accounted for other factors that specifically affected estimation of the total exposures, including counting AEs for the seniors only through the last game if they no longer practiced afterward. Using this method of calculating AEs, we were able to differentiate the exposures for initial and subsequent injuries, thus more accurately determining the rate of initial and subsequent injuries.

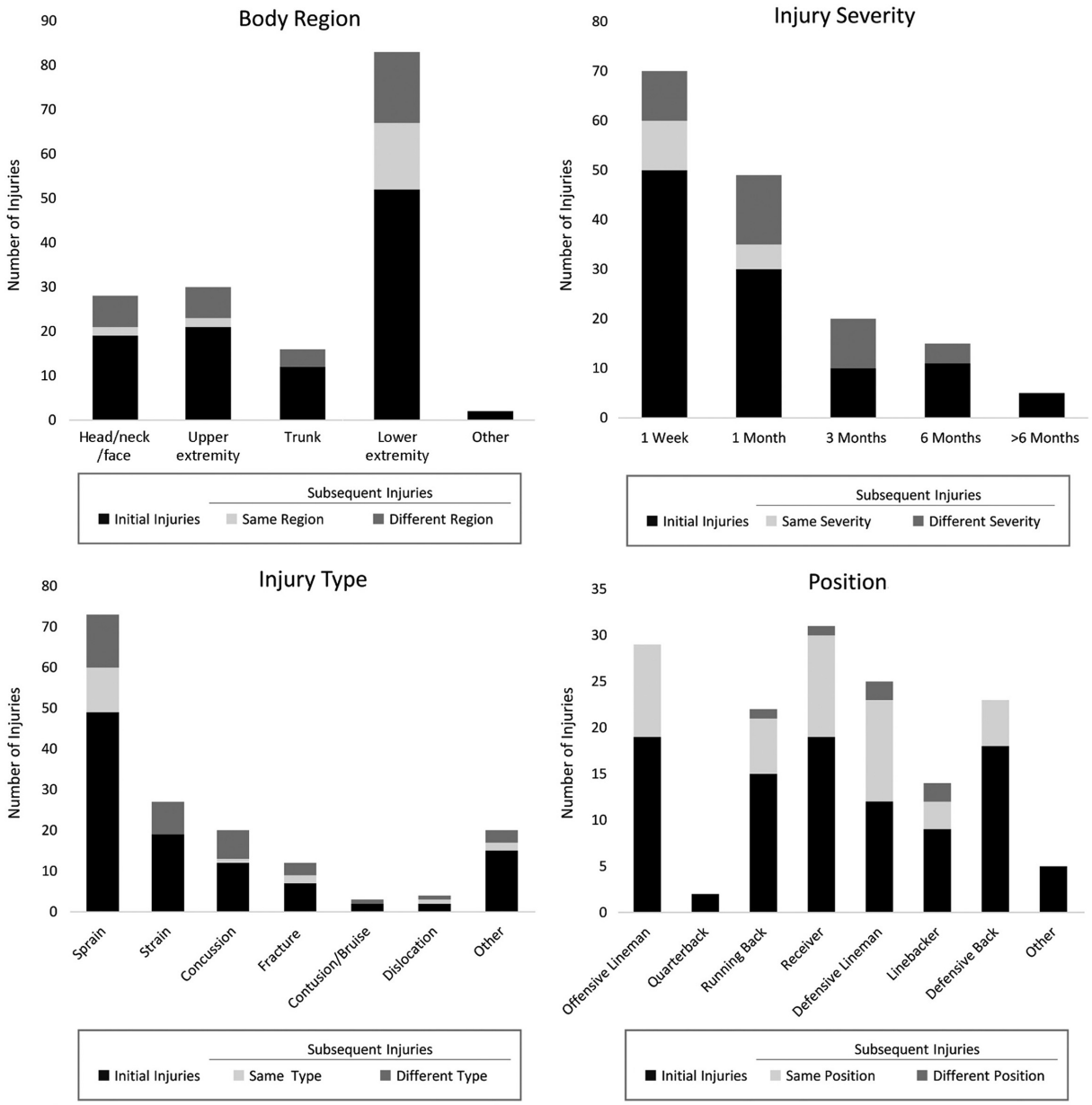

Figure 2. Characteristics of second injuries in relation to initial injuries: same vs different.

Consistent with previous findings that injured athletes have a greater risk to experience another injury,[7,14,20] this study found the rate of subsequent injuries to be nearly 1.5 times the rate of initial injuries, with an increased risk for each subsequent injury. Existing literature has reported several factors that could potentially contribute to the increased injury risk after an athlete's initial injury, including an individual player's patterns of risk-taking behaviors, premature return to play without full recovery from the initial injury, and altered biomechanics of joints after the initial injury.[4,8,14] While this study did not specifically examine each of these contributing factors, it examined several characteristics of athletes' initial injuries and their links to the proportions of athletes having sustained a second injury. The proportion of athletes who sustained a second injury tended to be higher among football players who had dislocations or fractures, who played in the defensive lineman position when injured, who sustained an injury to the lower extremities, or whose injury prevented return to play for 3 months. This approach linking initial and second injuries was also used in a previous study, in which functional testing was used along with history of injury to determine an athlete's risk.[15] Other studies have used a history of injury as proxy when assessing risk of subsequent injury.[14,18,20,23] The drawback with these studies, however, is that they lack defined timeframes for previous injuries and do not include exposures in their assessments. Future studies should also examine specific risk factors for subsequent

injury to determine which athletes may have a higher risk of subsequent injury. Once risk factors have been identified, clinicians, coaches, and athletes can be better informed about their decisions regarding return to play after an injury.

Another concerning trend in this study's findings is the high incidence of subsequent injuries to the head, neck, and face, which was nearly double the initial injury rate. Moreover, almost two-thirds (63%) of subsequent injuries to the head, neck, and face were concussions. The rate ratio (RR = 1.7; 95% CI, 0.9-3.5) for subsequent to initial concussions, although not statistically significant, was also high. These results are supported by existing findings on increased rates of concussions among athletes who have previously had a concussion.[6,9,16,17,27] Due to the evidence surrounding the short- and long-term negative effects of repeated concussions, including an increased risk of depression and a slower recovery time with second injury compared with first, these results further emphasize the importance of improving prevention efforts for subsequent injuries.[1,6,9,12,22]

The major limitations of this study resulted primarily from the study population. This study only included football teams at two Division I universities, and therefore results cannot be generalized to all collegiate athletes. Additionally, the small sample size prevented calculation of rates for specific risk factors. While analyses in this study addressed initial injury characteristics that result in second injuries, the small sample size also limited further analyses to determine how the characteristics of second injuries affect the likelihood of third injuries, and so on.[14] Furthermore, due to the way AEs were recorded in this database,[11] it was not possible to separate the players who actually played in competitions from those who attended competitions. It was also impossible to distinguish the subgroup of subsequently injured players who were more talented relative to their teammates and consequently played more during competitions, putting them at greater risk for both initial and subsequent injuries. The estimates of game injury rate in this study are likely an underestimate of the true rate. The injuries included in this study are likely to be more severe, as they require both "time loss at least 1 day" and "receiving medical attention" to be included. Finally, although this study defined the initial injury as the first injury during the study period, many of the athletes had a history of injury before enrolling in the study. Thus, the first injury reported may have actually been the athlete's second, third, or even higher order injury. Future studies with larger cohorts of athletes and more detailed information about athletes' history of injury would be ideal to appropriately power subanalyses to identify specific risk factors for subsequent injuries.

CONCLUSION

This study is unique in its application of a more accurate calculation of AEs and thus allows for a more truthful estimation of the risk of subsequent injury after an initial injury. Future research is necessary to better identify which factors contribute to the increased risk for subsequent injury among athletes with an initial injury.

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